Math 1220 Summer
Exam 2 Review

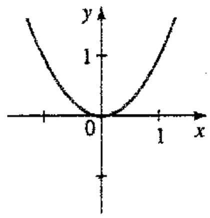
YOUR Name: ______________________________

7.2 Polynomial Functions

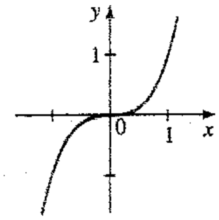
Power Functions

Monomial Polynomials,
$p(x) = x^n$

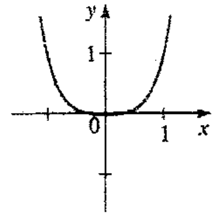
Transformation form,
$p(x) = a(x - h)^n + k$
where (h, k) is the "vertex" or the "center" point.



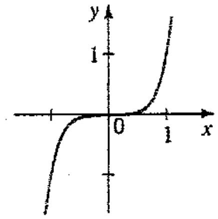
$y = x^2$



$y = x^3$



$y = x^4$



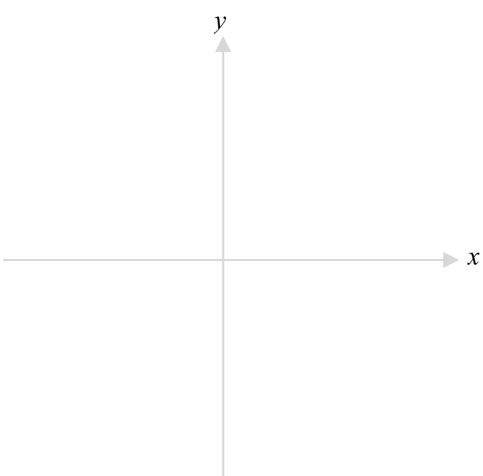
$y = x^5$

1. Graph each function. Find the intercepts and label them as numbers on the graph. Also indicate the "critical" point. Then write the domain and range in interval notation.

(a) $p(x) = \frac{1}{4}(x-2)^5 + 8$

D: ________________

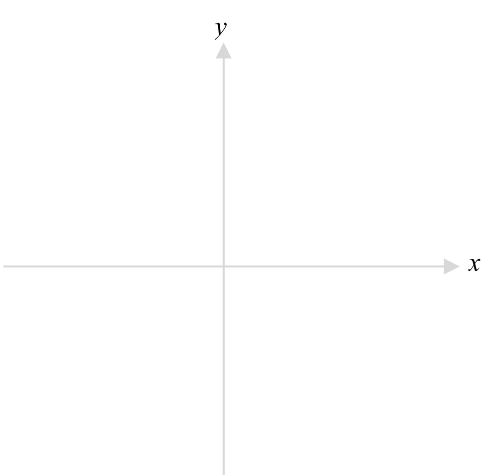
R: ________________



(b) $f(x) = -\frac{1}{4}(x+1)^4 + 4$

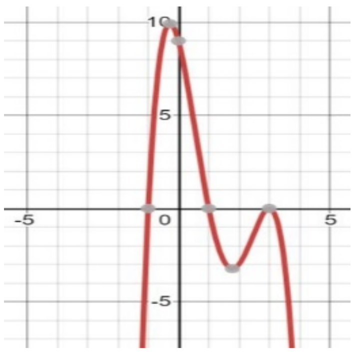
D: ________________

R: ________________



2. Determine the deg/sign of each polynomial. State each real zero and its multiplicity. Then write a feasible equation.

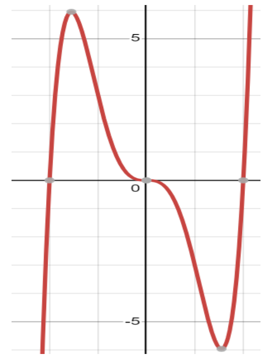
(a)



deg/sign: _________

zero ____ mult

(b)



deg/sign: _________

zero ______ mult

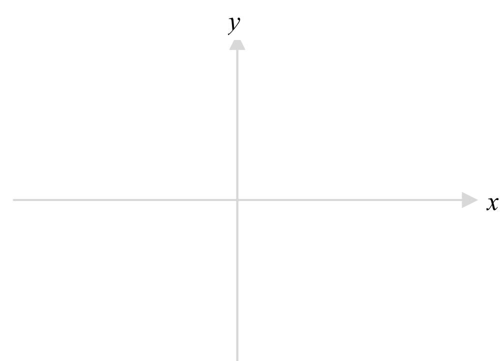
EQ:

EQ:

3. For each function, state the degree and sign of the leading coefficient, use arrows to show the end behavior, and then graph. Find all intercepts (factor if needed) and label them as numbers on the graph. Indicate any other "important" points if applicable.

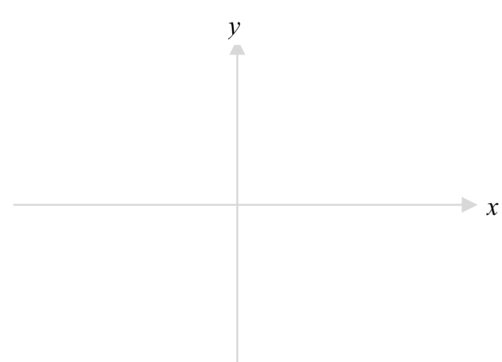
(a) $h(x) = -3x^2(x+2)(x-2)^3$

deg/sign of leading term: _________
end behavior:



(b) $g(x) = x^3 - 2x^2 - 3x$

deg/sign of leading term: _________
end behavior:



4. Find all intercepts, asymptotes and deleted points that exist for each function.

(a) $f(x) = \frac{8-4x}{2x+5}$ (b) $k(x) = \frac{2}{x^2+4}$ (c) $g(x) = \frac{x-2}{x^2-4}$

x-int: ______________
VA: ______________
y-int: ______________
HA: ______________
DP: ______________

x-int: ______________
VA: ______________
y-int: ______________
HA: ______________
DP: ______________

x-int: ______________
VA: ______________
y-int: ______________
HA: ______________
DP: ______________

5. Graph each function. Draw in and label all asymptotes, delete points and write all intercepts as numbers on the graph.

(a) $f(x) = \frac{2x+6}{x-3}$

VA(s): ______________
x-int: ______________
HA(s): ______________
y-int: ______________
DP: ______________

What are the asymptotes of $f^{-1}(x)$? ______________________________

(b) $g(x) = \frac{x-2}{x^2-4}$

VA(s): ______________
x-int: ______________
HA(s): ______________
y-int: ______________
DP: ______________

6. Find all intercepts and asymptotes that exist for each function and graph. You must justify why the function is above or below the x-axis over each interval determined by the x-intercepts and vertical asymptotes. Do not use a test value unless necessary!

(a) $f(x) = \frac{2x-3}{x^2+2x+1}$

VA(s): ______________

x-int: _______________

HA(s): ______________

y-int: ______________

DP: _____________

x-intervals:

Above or Below:

Justification:

(b) $g(x) = \frac{1-4x^2}{x^2+1}$

VA(s): ______________

x-int: _______________

HA(s): ______________

y-int: ______________

x-intervals:

Above or Below:

Justification:

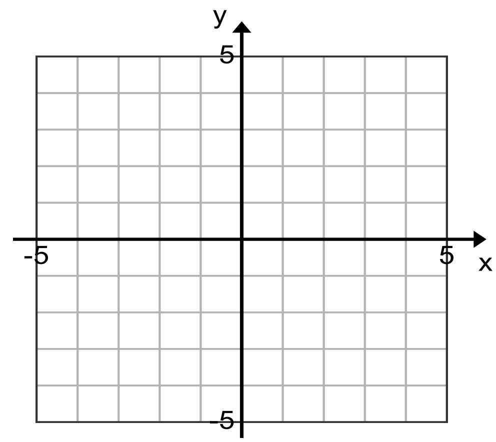
9.1 Inverse Functions

7. Let $f(x) = \frac{3-x}{2x+5}$. Find $f^{-1}(x)$ and state its domain and range.

9.2-9.3 Exponential & Logarithmic Functions

8. Fill in blanks and graph as indicated.

(a) Graph $f(x) = 2^x - 4$

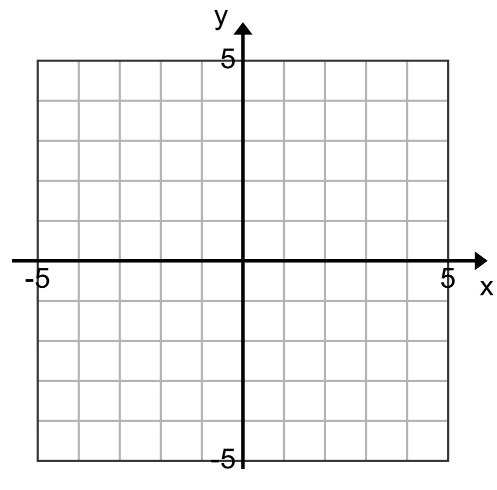


Domain: _______________

Range: ________________

Asymptote: ____________

(b) Graph $h(x) = \log_2(x+3)$



Domain: _______________

Range: ________________

Asymptote: ____________

9. If $h(x) = \log_2(x+3)$, find the following for $h^{-1}(x)$. Hint: h(x) is above.

Domain: _______________

Range: ________________

Asymptote: ____________

10. Convert each from exponential to logarithmic form, or vice-versa.

(a) $3^x = y$ (b) $e^x = 8$ (c) $\log_3 9 = 2$ (d) $\log x = 6$

(e) $A = B^C$ (f) $2^{x-1} = y$ (g) $\log_x Y = Z$ (h) $\ln(x-1) = y+2$

11. Find all intercepts of each function.

(a) $f(x) = e^{3-x} - 1$ (b) $g(x) = 4 - 2^{x+1}$

(c) $h(x) = \log_2(x+2) + 1$ (d) $k(x) = \ln(3-x) - 2$

12. Evaluate (a) $\log_2 32 =$ _____ (b) $\log_9 3 =$ _____ (c) $\log_8 \frac{1}{8} =$ _____ (d) $\log_6 \sqrt{6} =$ _____

(e) $\log_\pi 1 =$ _____ (f) $\log_p p =$ _____ (g) $\log_{27} 3 =$ _____ (h) $\ln(\frac{1}{e^2}) =$ _____

(i) $\log_3 3^8 =$ _____ (j) $(\log_3 3)^8 =$ _____ (k) $2\ln e^5 =$ _____ (l) $10^{\log 3\pi} =$ _____

(m) $(\log_4 2)^2 =$ _____ (n) $2e^{3\ln 2} =$ _____ (o) $\log(\ln e^{100}) =$ _____ (p) $\log_3(10^{2\log 3}) =$ _____

13. Combine into a single logarithm and simplify.

$\frac{1}{2}\log 16x^2 - \log x + 2\log(x-1)$

Separate $log_2 \frac{8x^5}{(x-1)^3}$ into simpler logs.

14. Solve (a) $3|\ln x| - 12 = 0$

(b) $5^{|x|} - 3 = 122$

(c) $\log x^2 = \log x$

(d) $50 = \frac{100}{1+3e^{-0.2t}}$

(e) $64^{2x-1} = 4(16^{x-3})$

(f) $10^{x+1} = 2^{3x-1}$

(g) $\log_4(x-2) = 1 - \log_4(x-5)$

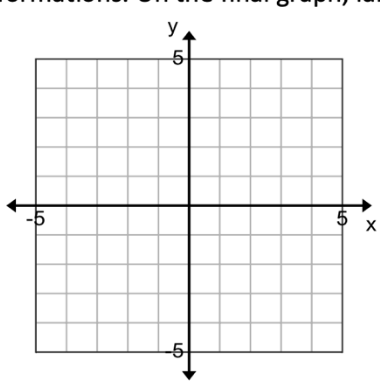
(h) $\ln 3x - 2\ln x = \ln 9$

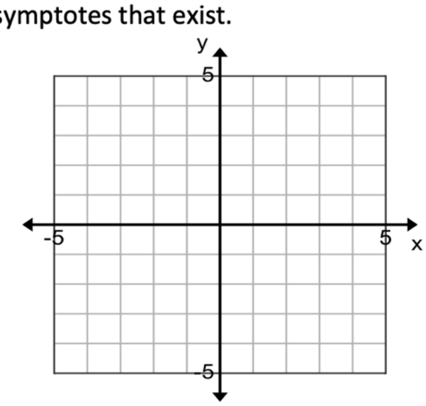
(i) $e^{2x} - 3e^x - 4 = 0$

(j) $e^{2x} = -9e^x$

Exam 1 Review Question

1. **Transformations**: Graph $y = 2\sqrt{3-x} + 1$. Use the 1st set of axes to sketch the parent function and work out the transformations. On the final graph, label all intercepts and asymptotes that exist.





2. Determine the domain for each function.

(a) $y = (x-1)(x+3)$

(b) $y = \frac{\sqrt{x+2}}{(x-1)(x+3)}$

(c) $y = \frac{(x-1)(x+3)}{\sqrt{x+2}}$

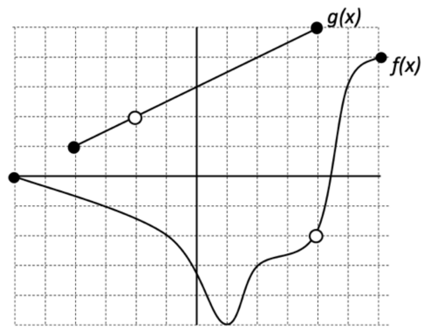
(d) $y = \frac{\log(x+3)}{x-1}$

3. Let $f(x) = \frac{1}{x-2}$ and $g(x) = \frac{1}{x}$. Find $(f \circ g)(x)$ and its domain.

4. **Example 2** Let *f* and *g* be the functions shown in the graph. Find the following.

(a) $(\frac{f}{g})(2)$=

(b) $(g \circ f)(6)$=



Quadratic Equations

Zero Product Property: If $mn = 0$, then $m = 0$ or $n = 0$

Square Root Property: If $x^2 = k$, then $x = \pm\sqrt{k}$

Quadratic Formula: If $ax^2 + bx + c = 0$, then $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

Solve (a) $2x^2 = 12x$

(b) $3(x-2)^2 = 96$

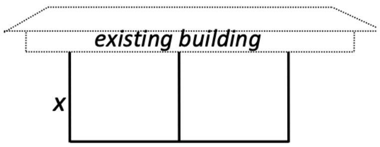
(c) $6y^2 - 17y = 3$

(d) $3x^2 = 8x - 1$

Quadratic Models

2. A gardener has 120 meters of fencing to enclose two adjacent rectangular growing plots. One side is to be against a building as shown.

(a) If x represents the width of the plot, express its area A(x) in terms of x.



(b) Determine the dimensions of the rectangle that will make the area a maximum. What is the max growing area?

4. The number of bacteria in a culture is modeled by the function $P(t) = 500e^{0.4t}$ where t is measured in hours.

(a) The initial number of bacteria is ________.

(b) The relative growth rate of this bacterium population (as a percentage) is _________.

(c) After how many hours will the population reach 10,000?